

Education

Cornell University | College of Arts & Science

Expected Graduation: May 2028

Computer Science, Statistics

GPA: 3.78

Relevant Coursework: Computer System Organization (CS 3410), Algorithms (CS 4820), Machine Learning (CS 3780), Data Structures and Functional Programming (CS 3110), Discrete Math (CS 2800), Probability Models and Inference (STSCI 3080), Statistics II (STSCI 3200), Categorical Data Analysis (STSCI 4110), Linear Algebra, Multivariable Calculus, Differential Equations

Work

Cornell University, Department of Civil Engineering

Jun 2026 – Present

Research Assistant

- Developed a particle-based data assimilation pipeline to estimate unobserved irrigation inputs by integrating satellite observations with land-surface model outputs.
- Implemented likelihood-based particle weighting, resampling, and posterior estimation across multi-day inference windows using Python, xarray, pandas, and NetCDF.
- Built preprocessing and calibration workflows to align heterogeneous spatiotemporal datasets across model grids, satellite observations, and watershed boundaries.
- Designed adaptive filtering methods using rainfall, seasonal constraints, and satellite data-quality checks to remove noisy inputs

University of Illinois, Urbana-Champaign, College of Education

Sep 2023 – May 2024

Research Intern

- Developed a customizable map application for Illinois's I3 project under Professor Asif Wilson, enabling K-12 teachers to access 1000+ cultural and historical resources for inquiry-driven social studies curricula.
- Automated resource classification and enhanced tag precision using ChatGPT-powered pre/post processing.

Experience

AlgoVerse LLM Research Group

May 2026 – Present

Researcher

- Investigating whether linear probes can detect representational drift in long-horizon LLM forecasting agents before forecast outputs become miscalibrated or unstable.
- Assembled time-aligned forecasting datasets by combining prediction-market questions, historical market prices, order-book data, and timestamped news streams.
- Developed behavioral metrics for anchoring, forecast instability, sample dispersion, and directional update consistency across repeated forecasting trials.

Cornell Geo Data

Nov 2025 – Present

Student Researcher

- Developed a correctional ML model to improve forecasting of lake-effect snow near the Finger Lakes by integrating HRRR numerical weather predictions with ground-truth sensor data and optimizing the model with a GAN-based approach.
- Utilized Autodesk Inventor to design a dimensioned schematic for a planned water buoy deployment.
- Calibrated sensors, processed raw field data, and researched local hydrological and atmospheric models.

Harvard Institute of Quantitative Social Sciences

Jun 2024 – Aug 2024

Research Intern

- Applied difference-in-differences regressions to assess encampment-ban policy impacts across urban counties using HUD homeless data and Census statistics.
- Identified implementation variables—penalties, enforcement constraints, shelter access—as key determinants of homelessness outcomes.
- Published findings in Journal of Student Research: [Why Encampment Bans Don't Reduce Homelessness](#).

Aspiring Scholars Directed Research Program

Jan 2023 – Jun 2024

Research Intern

- Developed 5 custom Constitutional AI principles to reduce political bias in ChatGPT-3.5, testing across 300+ expert-approved political prompts.
- Published results in JEI, and presented at Southern California's Conference for Undergraduate Research: [Analyzing Constitutional AI Principles for Politically-Biased Responses](#).

Skills

Languages: C/C++, Python, Java, R, JavaScript, SQL, OCaml, LaTeX

Data & ML: pandas, NumPy, scikit-learn, PyTorch, TensorFlow, matplotlib, xarray, NetCDF

Tools: Git, Jupyter, Linux/Unix, Google Earth Pro